

Technology #161

incorporate a multiphase nanocomposite
oxygen■scavenging additive, analogous to the
MNCP, into the pouch■cell electrolyte or
separator to actively oxidize residual oxygen
during formation cycling, thereby preventing
gas evolution

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-09-02

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

incorporate a multiphase nanocomposite oxygen■scavenging additive, analogous to the MNCP, into the pouch■cell electrolyte or separator to actively oxidize residual oxygen during formation cycling, thereby preventing gas evolution

WHY DOES IT MATTER?

Documented problem 'gas-induced swelling' concerning lithium-ion pouch cell. User-submitted engineering problem (session input): "Why do lithium-ion pouch cells blister and swell during formation cycling before any external abuse, and how can we suppress gas-induced swelling without adding a vent?". ROUTING: nhtsa_complaints: not queried — no vehicle (make|model|year) extracted from the problem; this source answers vehicle-parameterized questions only. [nhtsa_complaints] retrieval status NOT_QUERIED_GRAMMAR (no vehicle extracted; source grammar is make|model|year (routing decision, not absence)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] 10 records retrieved live (2026-2026). Illustrative records: 'Understanding the thermal stability of high-energy density alloy anodes is critical for the safe deployment of sodium-ion batteries (SIBs). Here, accelerating rate calorimetry (ARC), post-mortem chara'; 'Formation of robust solid/cathode electrolyte interphases (S/CEI) is vital for long-term stability and high-performance operation of lithium-ion batteries (LIBs), particularly under high voltage regim'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in energy_harvesting. 10-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 2 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 10-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: run parallel formation cycles on cells with and without the oxygen scavenging additive, monitor internal pressure rise and pouch thickness.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: transduced power vs source excitation sweep. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: S_{diff} is material- and temperature-dependent — source the value for the actual couple; output scales with the MAINTAINED difference, not with ambient temperature alone; coupling coefficients are frequency-, load- and temperature-dependent; quoted values are not design values until measured in situ; real conditioning chains never achieve the theoretical maximum — measure the achieved fraction; ENH-001 (Seebeck effect (thermoelectric voltage)) is applicable only under: couple materials characterized; maintained temperature difference across the junctions.; ENH-002 (Piezoelectric constitutive relation (charge form)) is applicable only under: small-signal linear regime below depoling limits.; ENH-003 (Maximum power transfer to a load) is applicable only under: linear Thevenin-equivalent source; matched load.; Model input variables requiring sourced values before any numeric result: S_{diff} (V/K), dT (K), d_T (C/N), T_{stress} (Pa), eps_T (F/m), E (V/m), V_{oc} (V), R_{int} (ohm) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: V (V), D (C/m²), P_{max} (W) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): ambient source magnitude and spectrum [V_{oc} , V] value_status=UNKNOWN; transduction coefficient [S_{diff} , V/K] value_status=UNKNOWN; conditioning efficiency [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; storage capacity and leakage [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; load duty profile [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; startup threshold power [P_{max} , W] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

2 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: oxygen oxidation (as a chemical reaction that consumes oxygen and does not generate gas)

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'gas-induced swelling' concerning lithium-ion pouch cell. User-submitted engineering problem (session input): "Why do lithium-ion pouch cells blister and swell during formation cycling before any external abuse, and how can we suppress gas-induced swelling without adding a vent?". ROUTING: nhtsa_complaints: not queried — no vehicle (make|model|year) extracted from the problem; this source answers vehicle-parameterized questions only. [nhtsa_complaints] retrieval status NOT_QUERIED_GRAMMAR (no vehicle extracted; source grammar is make|model|year (routing decision, not absence)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] 10 records retrieved live (2026-2026). Illustrative records: 'Understanding the thermal stability of high-energy density alloy anodes is critical for the safe deployment of sodium-ion batteries (SIBs). Here, accelerating rate calorimetry (ARC), post-mortem chara'; 'Formation of robust solid/cathode electrolyte interphases (S/CEI) is vital for long-term stability and high-performance operation of lithium-ion batteries (LIBs), particularly under high voltage regim'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: energy_harvesting

Disciplines: energy conversion, power electronics, materials science, mechanical engineering

Architecture: PROPOSED architecture (no physical system exists): ambient source characterization element; transduction stage (piezo / thermo / inductive); power conditioning and storage buffer; load interface + duty-cycle governor — realizing: incorporate a multiphase nanocomposite oxygen■scavenging additive, analogous to the MNCP, into the pouch■cell electrolyte or separator to actively oxidize residual oxygen during formation cycling, thereby preventing gas evolution

Subsystems:

?: ambient source characterization element - ?

?: transduction stage (piezo / thermo / inductive) - ?

?: power conditioning and storage buffer - ?

?: load interface + duty-cycle governor - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: incorporate a multiphase nanocomposite oxygen■scavenging additive, analogous to the MNCP, into the pouch■cell electrolyte or separator to actively oxidize residual oxygen during formation cycling, thereby preventing gas evolution

Key Physics: Energy harvesting / power conversion governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Energy harvesting / power conversion governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

ENH-001: Seebeck effect (thermoelectric voltage). Expression: $V = S_{diff} * dT$. Applicability to this invention: CONDITIONAL; holds under: couple materials characterized; maintained temperature difference across the junctions. Assumptions: S_{diff} is material- and temperature-dependent — source the value for the actual couple; output scales with the MAINTAINED difference, not with ambient temperature alone. Engaged variables: NOT ESTABLISHED.

ENH-002: Piezoelectric constitutive relation (charge form). Expression: $D = d_T * T_{stress} + \epsilon_{sT} * E$. Applicability to this invention: CONDITIONAL; holds under: small-signal linear regime below depoling limits. Assumptions: coupling coefficients are frequency-, load- and temperature-dependent; quoted values are not design values until measured in situ. Engaged variables: NOT ESTABLISHED.

ENH-003: Maximum power transfer to a load. Expression: $P_{max} = V_{oc}^2 / (4 * R_{int})$. Applicability to this invention: CONDITIONAL; holds under: linear Thevenin-equivalent source; matched load. Assumptions: real conditioning chains never achieve the theoretical maximum — measure the achieved fraction. Engaged variables: NOT ESTABLISHED.

Assumptions:

- S_{diff} is material- and temperature-dependent — source the value for the actual couple
- output scales with the MAINTAINED difference, not with ambient temperature alone
- coupling coefficients are frequency-, load- and temperature-dependent; quoted values are not design values until measured in situ
- real conditioning chains never achieve the theoretical maximum — measure the achieved fraction
- ENH-001 (Seebeck effect (thermoelectric voltage)) is applicable only under: couple materials characterized; maintained temperature difference across the junctions.

- ENH-002 (Piezoelectric constitutive relation (charge form)) is applicable only under: small-signal linear regime below depoling limits.
- ENH-003 (Maximum power transfer to a load) is applicable only under: linear Thevenin-equivalent source; matched load.
- Model input variables requiring sourced values before any numeric result: S_{diff} (V/K), dT (K), d_T (C/N), T_{stress} (Pa), ϵ_{ps_T} (F/m), E (V/m), V_{oc} (V), R_{int} (ohm) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: V (V), D (C/m²), P_{max} (W) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): ambient source magnitude and spectrum [V_{oc} , V] value_status=UNKNOWN; transduction coefficient [S_{diff} , V/K] value_status=UNKNOWN; conditioning efficiency [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; storage capacity and leakage [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; load duty profile [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; startup threshold power [P_{max} , W] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'gas-induced swelling', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'harvested power below startup threshold', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'transducer depoling/aging loss', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'storage leakage draining buffer between events', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'load transient collapsing supply rail', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'encapsulation stress cracking transducer', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	suppress swelling without adding a vent while preserving cell performance	SOURCE_FACT
DI-002	Mechanism driver (proposed)	oxygen oxidation (as a chemical reaction that consumes oxygen and does not release gas)	MODELED
DI-003	Expected effect target	suppression of gas-induced swelling and blistering during formation without need for a vent, while maintaining lithium	MODELED
DI-004	Domain design input pattern: available application	NOT ESTABLISHED (candidate pattern; confirm against engineering profile)	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: load power application	NOT ESTABLISHED (candidate pattern; confirm against engineering profile)	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: recharge application	NOT ESTABLISHED (candidate pattern; confirm against engineering profile)	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance standards	1500000 db (electrolyte) and fully unknown (electrolyte) is never inferred	ENGINEERING_PROPOSED
DI-008	Custodied observation: Flexible Sensing	https://pubs.rsc.org/en/article/2014/5/1471g Motoring Materials 2013, 33, 5972-5981	666752d105a0cdd7b984a0
DI-009	Custodied observation: Flash up cycling	https://arxiv.org/abs/2014.11.02774 in today's crash pre-99ed1096704f9e5e407e5af8db8007eab1fb57dba	
DI-010	Custodied observation: A Self-Describing	https://pubs.rsc.org/en/article/2014/17/43758 Whemingash 2013, 101, 13, 11639-11643	629615444f93e9a74c1605
DI-011	Custodied observation: Advanced Technology	https://pubs.rsc.org/en/article/2014/17/43758 Battery Test 95, 591-597	7a7e201f4359e437cfdc3b8472b2632161ead
DI-012	Custodied observation: Multiphase	https://pubs.rsc.org/en/article/2014/17/43758 c of the in the 5000000044550054-aa1056f482c5ccad0c4576db744	

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	ambient source characterization element realizing incorporation of multiphase nanocomposite oxygen scavenging additive, analogous to the MNCP, into the	CONCEPTUAL	not detailed as oxygen scavenging additive, analogous to the MNCP, into the
DO-002	transduction stage (piezo / thermo / inductive) realizing incorporation of multiphase nanocomposite oxygen scavenging additive, analogous to the MNCP, into the	CONCEPTUAL	not detailed as oxygen scavenging additive, analogous to the MNCP, into the
DO-003	power conditioning and storage buffer realizing incorporation of multiphase nanocomposite oxygen scavenging additive, analogous to the MNCP, into the	CONCEPTUAL	not detailed as oxygen scavenging additive, analogous to the MNCP, into the
DO-004	load interface + duty-cycle governor realizing incorporation of multiphase nanocomposite oxygen scavenging additive, analogous to the MNCP, into the po	CONCEPTUAL	not detailed as oxygen scavenging additive, analogous to the MNCP, into the po
DO-005	ambient source characterization element interfaces with transduction stage (piezo / thermo / inductive) (signal/energy/materials)	CONCEPTUAL	not detailed as oxygen scavenging additive, analogous to the MNCP, into the po
DO-006	transduction stage (piezo / thermo / inductive) interfaces with power conditioning and storage buffer (signal/energy/materials)	CONCEPTUAL	not detailed as oxygen scavenging additive, analogous to the MNCP, into the po
DO-007	power conditioning and storage buffer interfaces with load interface + duty-cycle governor (signal/energy/materials)	CONCEPTUAL	not detailed as oxygen scavenging additive, analogous to the MNCP, into the po
DO-008	Device boundaries: patient/fluid inlet and outlet terminal control/data points required by selection of multiphase nanocomposite oxygen scavenging	CONCEPTUAL	not detailed as oxygen scavenging additive, analogous to the MNCP, into the po
DO-009	Flow/force/signal path geometry realizing the mechanism to incorporate a multiphase nanocomposite oxygen scavenging additive, analogous to the MNCP, into the	UNKNOWN	mechanism to incorporate a multiphase nanocomposite oxygen scavenging additive, analogous to the MNCP, into the
DO-010	Operating range for ambient source magnitude and spectrum	UNKNOWN	mechanism to incorporate a multiphase nanocomposite oxygen scavenging additive, analogous to the MNCP, into the
DO-011	Operating range for transduction coefficient within which the mechanism holds its stated behavior	UNKNOWN	mechanism to incorporate a multiphase nanocomposite oxygen scavenging additive, analogous to the MNCP, into the
DO-012	Operating range for conditioning efficiency within which the mechanism holds its stated behavior	UNKNOWN	mechanism to incorporate a multiphase nanocomposite oxygen scavenging additive, analogous to the MNCP, into the
DO-013	Operating range for storage capacity and leakage with the mechanism holds its stated behavior	UNKNOWN	mechanism to incorporate a multiphase nanocomposite oxygen scavenging additive, analogous to the MNCP, into the
DO-014	Operating range for load duty profile within which the mechanism holds its stated behavior	UNKNOWN	mechanism to incorporate a multiphase nanocomposite oxygen scavenging additive, analogous to the MNCP, into the
DO-015	Operating range for startup threshold power within which the mechanism holds its stated behavior	UNKNOWN	mechanism to incorporate a multiphase nanocomposite oxygen scavenging additive, analogous to the MNCP, into the
DO-016	Sensing -> decision -> actuation loop maintaining the proposed operating point	PROPOSED	sensing capability selection; decision to deposit (oxygen scavenging additive)
DO-017	Information path: sensing element -> decision/monitor control	CONCEPTUAL	act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in
DO-018	Fixture enabling: transduced power vs source excitation	UNKNOWN	rig concept; fixed rig concept; fixed rig concept; fixed rig concept; fixed rig concept
DO-019	Fixture enabling: conditioning-chain conversion efficiency	UNKNOWN	rig concept; fixed rig concept; fixed rig concept; fixed rig concept; fixed rig concept
DO-020	Fixture enabling: duty-cycle endurance under worst-case	UNKNOWN	rig concept; fixed rig concept; fixed rig concept; fixed rig concept; fixed rig concept

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: power vs excitation sweep characterization	UNKNOWN	Unknown concept name; budget specification; unknown selection; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
ambient source magnitude	UNKNOWN (no sourced value)	V	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
transduction coefficient	UNKNOWN (no sourced value)	V/K	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
conditioning efficiency	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
storage capacity and leakage	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
load duty profile	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
startup threshold power	UNKNOWN (no sourced value)	W	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
gas-induced swelling	Fails by gas-induced swelling	Get the device to NOT GET ABOVE 652 (equivalent design) UNKNOWN the operating conditions	UNKNOWN	candidate failure mode (Artec)
harvested power below startup threshold	Fails by harvested power below startup threshold	at the 5005, combine ing that it will be able to meet the required power	UNKNOWN	candidate failure mode (Aing)
transducer depoling/aging	Fails by transducer depoling/aging	limiting time detating, aging and preheating; stress postively depoled degraded	UNKNOWN	candidate failure mode (Artec)
storage leakage draining	Fails by storage leakage draining	draining to be below levels, which self-discharge and unknown how connected	UNKNOWN	candidate failure mode (Artec)
load transient collapse	Fails by load transient collapse	storage supply float transients, which exceeds as scheduled	UNKNOWN	candidate failure mode (Artec)
encapsulation stress cracking	Fails by encapsulation stress cracking	encapsulation material power is leakage and unknown this results	UNKNOWN	candidate failure mode (Artec)

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
gas-induced swelling	Fails by gas-induced swelling: the device degrades due to NO ₂ ESTABLISHED (defined after verification)	design change: NO ₂ ESTABLISHED (defined after verification)	design change: NO ₂ ESTABLISHED (defined after verification)
harvested power below start-up threshold	Fails by electrostatic power below start-up threshold: electrostatic abuse site conditions: NO ₂ ESTABLISHED (defined after verification)	design change: NO ₂ ESTABLISHED (defined after verification)	design change: NO ₂ ESTABLISHED (defined after verification)
transducer depoling/aging	Fails by transducer depoling/aging loss of field temperature and aging: aging: NO ₂ ESTABLISHED (defined after verification)	design change: NO ₂ ESTABLISHED (defined after verification)	design change: NO ₂ ESTABLISHED (defined after verification)
storage leakage draining	Fails by storage leakage draining buffer storage: leakage: NO ₂ ESTABLISHED (defined after verification)	design change: NO ₂ ESTABLISHED (defined after verification)	design change: NO ₂ ESTABLISHED (defined after verification)
load transient collapsing	Fails by load transient collapsing supply storage: leakage: NO ₂ ESTABLISHED (defined after verification)	design change: NO ₂ ESTABLISHED (defined after verification)	design change: NO ₂ ESTABLISHED (defined after verification)
encapsulation stress cracking	Fails by encapsulation stress cracking transducer encapsulation materials: leakage: NO ₂ ESTABLISHED (defined after verification)	design change: NO ₂ ESTABLISHED (defined after verification)	design change: NO ₂ ESTABLISHED (defined after verification)

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	run parallel formation cycles on cells with parallel formation cycles	run parallel formation cycles on cells with parallel formation cycles	run parallel formation cycles on cells with parallel formation cycles	run parallel formation cycles on cells with parallel formation cycles
VF-002	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test
VF-003	transduced power vs source excitation	transduced power vs source excitation	transduced power vs source excitation	transduced power vs source excitation
VF-004	conditioning-chain conversion efficiency	conditioning-chain conversion efficiency	conditioning-chain conversion efficiency	conditioning-chain conversion efficiency
VF-005	power vs excitation sweep with startup	power vs excitation sweep with startup	power vs excitation sweep with startup	power vs excitation sweep with startup
VF-006	energy-balance measurement over the energy-balance measurement	energy-balance measurement over the energy-balance measurement	energy-balance measurement over the energy-balance measurement	energy-balance measurement over the energy-balance measurement
VF-007	rail-ripple measurement under worst-case	rail-ripple measurement under worst-case	rail-ripple measurement under worst-case	rail-ripple measurement under worst-case
VF-008	acoustic micro-imaging plus output	acoustic micro-imaging plus output	acoustic micro-imaging plus output	acoustic micro-imaging plus output

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	PZT / AlN piezoelectric stack	implantable energy harvester practice	True	ENGINEERING_PROPOSED
domain candidate material	BiTe thermoelectric couple	body-heat harvesting practice	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
PZT / AlN piezoelectric stack	domain candidate material	NOT ESTABLISHED	NOT ESTABLISHED	PZT / AlN piezoelectric stack	NOT ESTABLISHED	NOT ESTABLISHED	Reliability verification required
BiTe thermoelectric couple	domain candidate material	NOT ESTABLISHED	NOT ESTABLISHED	BiTe thermoelectric couple	NOT ESTABLISHED	NOT ESTABLISHED	Reliability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- thin-film transducer deposition and dicing: domain registry manufacturing route - candidate only; no process qualification exists
- hermetic electronics encapsulation: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

Source 2:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-002', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter ambient source magnitude and spectrum [V_oc V]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter transduction coefficient [S_diff V/K]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter conditioning efficiency [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter storage capacity and leakage [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter load duty profile [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter startup threshold power [P_max W]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}

- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-005', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-005'}
- {'unknown': 'acceptance criterion for verification VF-006', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-006'}
- {'unknown': 'acceptance criterion for verification VF-007', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-007'}
- {'unknown': 'acceptance criterion for verification VF-008', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-008'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	maintaining the proposed physical verification	100 Hz frequency response	NOT ESTABLISHED	NOT ESTABLISHED	NOT ESTABLISHED
WP-02	constraint-compliance envelope article	per method	constraint-compliance	NOT ESTABLISHED	constraint-compliance	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	run parallel formation	NOT ESTABLISHED	VF-001 verification	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment:	NOT ESTABLISHED	VF-002 verification	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	transduced power vs	NOT ESTABLISHED	VF-003 verification	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	conditioning-chain con	NOT ESTABLISHED	VF-004 verification	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	power vs excitation sw	NOT ESTABLISHED	VF-005 verification	NOT ESTABLISHED (to be quoted)
WP-08	article for VF-006	per method	energy-balance measu	NOT ESTABLISHED	VF-006 verification	NOT ESTABLISHED (to be quoted)
WP-09	article for VF-007	per method	rail-ripple measureme	NOT ESTABLISHED	VF-007 verification	NOT ESTABLISHED (to be quoted)
WP-10	article for VF-008	per method	acoustic micro-imagi	NOT ESTABLISHED	VF-008 verification	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 10-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.